

Deliverable: Project Analysis & Methodology

Approach: Data was sourced from EMPRES-i and filtered for swine-related disease impacts. The data was split into 70% Training, 15% Validation, and 15% Test sets to ensure unbiased evaluation[cite: 1].

Best Performing Model: The **Random Forest** performed best[cite: 1].

Why: Tree-based ensemble methods like Random Forest and Gradient Boosting effectively handle the geographic non-linearity of disease outbreaks[cite: 1]. By optimizing for F1-Score, the model balances the need to identify all outbreaks (Recall) while maintaining high diagnostic accuracy (Precision)[cite: 1].

